

Surface grafting of styrene on polypropylene by argon plasma and its adsorption and regeneration of BTX

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ABSTRACT: Active macromolecular free radicals were generated on polypropylene (PP) fiber surfaces by argon plasma irradiation, and surface-modified PP fibers (PP-g-St fibers) were prepared by *in situ* grafting reaction of styrene monomers (St). The prepared samples were characterized by Fourier transform infrared, NMR, X-ray photoelectron spectroscopy, scanning electron microscopy, and thermogravimetric analysis. Effects of reaction parameters on grafting percentage were studied, and adsorption capacities of PP-g-St fibers for benzene, toluene, and xylene (BTX) were evaluated. Regeneration adsorption efficiencies after adsorption of pure BTX and BTX emulsion and solution in water were explored. The results indicated that, using pure St as the monomer, the optimum input power, irradiation time, and grafting reaction time are 90 W, 3 min, and 3 h, respectively, and the grafting percentage of St reached 5.7% when pure St was used. The characterization results demonstrated that St was grafted onto the surface of the PP fibers. Compared to pristine PP fibers, the adsorption capacities of PP-g-St fibers toward toluene and xylene emulsions and solutions in water increased. In addition, regeneration adsorption efficiencies of modified fibers remained >90% after six cycles of regeneration adsorption experiments, which showed excellent regeneration ability. © 2018 Wiley Periodicals, Inc. *J. Appl. Polym. Sci.* **2018**, 136, 46171.

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INTRODUCTION

Benzene, toluene, and xylene (BTX) are typical pollutants and generally form emulsions in wastewaters when discharged either from industrial plants or during transportation and storage. Due to their low degradability and highly poisonous nature, they pose a great threat to the environment and to human health if discharged without treatment.^{1,2} The main methods to date for treatment of BTX in wastewaters include biodegradation,³ thermal oxidation, catalytic oxidation, condensation and separation, membrane separation, and adsorption.⁴ Adsorption is considered to be one of the most efficient methods for treating organic pollutants in wastewaters, due to several advantages, such as simple operation, no secondary pollution, fast processing, and renewability of BTX.^{5–7} The adsorption materials can be either inorganic (such as diatomite, attapulgite, or volcanics), natural organic fibers (such as cotton, linen, and wool), or

synthetic organic in nature (such as macroporous resin, polymeric fibers, and polyurethane foams).⁸

Polypropylene (PP) fiber, as a kind of synthetic organic adsorption material, has been widely used for the treatment of oily wastewaters with the merits of large specific surface area, good hydrophobic and lipophilic properties, good rigidity, and stable physicochemical properties.^{9,10} However, the affinity between PP fibers and BTX is low because of the difference in their solubility parameters, which limits the adsorption of BTX, especially in the case of BTX emulsion and solution in water. Hence, it is necessary to modify the PP fiber surface to improve its affinity to BTX. Melt grafting, solid grafting, liquid-phase grafting, and suspension grafting are the main modification methods that are practically useful,^{11,12} but these methods have some drawbacks, such as severe degradation of PP, high cost, and nonfibrous nature of the modified product, and they are also

environmentally harmful. Plasma-induced grafting, an excellent technique for surface modification, has many outstanding features, like mild processing conditions and environmentally friendly nature, and it also causes no damage to the internal structure of the material.^{13–16} The principle of this technique is to produce active sites (free radicals) by bombarding the material surface with energetic ionizing particles, where an *in situ* grafting reaction of monomers takes place.¹⁷

Polystyrene (PS) has a large number of styrene (St) units in its molecular chain, which is similar to the molecular structure of BTX, and their solubility parameters are also close.¹⁸ So, PS has excellent adsorption and occlusion for BTX. However, the swelling and dissolution of PS in BTX cause deformation and loss of mechanical properties of the PS fiber. Moreover, it is difficult to separate the BTX occluded inside the PS fibers, and its regeneration is difficult. Several studies have reported the grafting of St on PP through photochemical,¹⁹ gamma-ray irradiation,²⁰ plasma treatment,²¹ supercritical CO₂,²² electron beam treatment,^{23,24} and solid-state grafting.²⁵ The mechanical property improvements of the modified PP materials were mainly investigated. Generally, irradiation induced the production of radicals on the solid polymer, and then initiators are needed for vinyl graft polymerization.^{21–23}

In this paper, by combining the advantages of PP and PS for adsorption of BTX, active macromolecular free radicals were generated on a PP fiber matrix using argon (Ar) plasma irradiation. No added initiator or other additives were used in this study. The *in situ* grafting reaction of St monomers occurred, and PP-g-St fibers were formed. The effects of reaction parameters input power, irradiation time, grafting reaction time, and concentration of St on grafting percentage were studied. Adsorption capacities of PP-g-St fibers for BTX emulsion and solution in water were evaluated, and their maximum adsorption capacities for pure BTX were measured as well. Finally, regeneration methods and regeneration adsorption efficiencies after maximum adsorptions were explored.

EXPERIMENTAL

Materials and Instruments

Homemade melt-blown PP fibers (fiber diameters, 2–10 μm ; specific area, 1.50 m^2/g) were used as the backbone.²⁶ St (CP grade) was distilled under reduced pressure to remove polymerization inhibitor prior to use. Benzene, toluene, xylene, acetone, and anhydrous sodium sulfate were all of analytical grade. Sodium dodecyl benzene sulfonate (SDBS) was of CP grade, carbon tetrachloride was specially produced for environmental protection, and petroleum ether was of industrial grade. The plasma regenerator, including a 500 W radio frequency power source and 2000 W radio frequency matcher, were all provided by the Microelectronics Institute of Chinese Academy of Sciences (Beijing, China).

Preparation of PP-g-St Fibers

The preparation process is shown in Figure 1 and can be described in the following steps. PP fibers were immersed in acetone for 2 h and dried to a constant weight at 50 $^{\circ}\text{C}$ before use. The PP fibers (10) were sampled and weighed and were

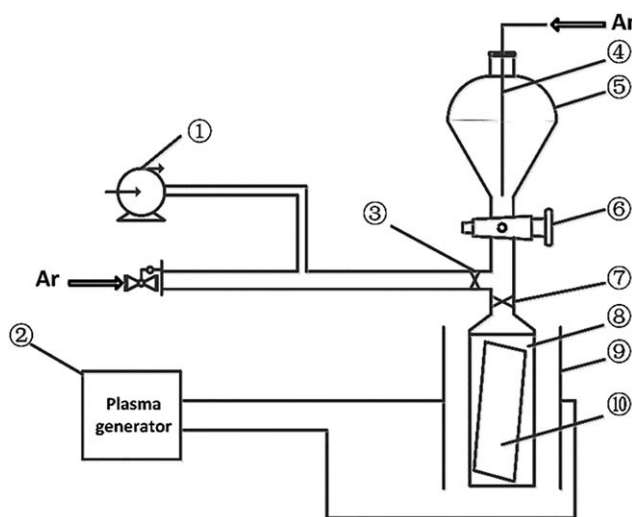
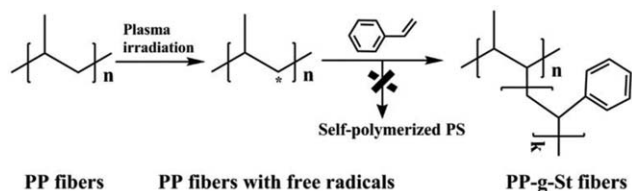


Figure 1. Setup for the preparation of PP-g-St fibers by plasma-induced grafting (1: vacuum pump, 2: plasma generator, 3: valve A, 4: metal gas inlet pipe, 5: separating funnel for St, 6: valve B, 7: valve C, 8: reactor, 9: electrodes, 10: fiber sample).

then transferred to the reactor (8). The reactor was placed between electrodes (9) of the plasma regenerator (2). The solution of St monomers was taken in the separating funnel (5), and Ar gas was bubbled into the solution to eliminate air when valve B (6) was closed. At the same time, valves A (3) and C (7) were then opened and air was removed until the pressure reached 5 Pa. The air in the reactor (8) was replaced with Ar gas three times. Ar gas was again pumped in and evacuated to 5 Pa. Valves A (3) and C (7) were closed, and the plasma generator (2) was switched on to irradiate the fibers under set plasma conditions. The plasma generator (2) was turned off, and the valves B (6) and C (7) were opened to guide the solution of St monomers into the reactor (8). Valve B (6) should be closed after the fiber sample (10) was totally immersed in the St monomer solution. This was the beginning of the grafting reaction. The grafting reaction ended when valve B (6) was opened again to introduce air. After that, fiber samples (10) were taken out and extracted with acetone for 5 h to get rid of the remaining unreacted St and the self-polymerized PS. Finally, the fiber samples (10) were dried at 50 $^{\circ}\text{C}$ until they attained a constant weight.

The mechanism for the preparation of PP-g-St fibers by plasma-induced grafting is shown in Scheme 1. During plasma treatment and the monomer grafting process, degradation and rearrangement of PP radicals may occur, as well as



Scheme 1. Reaction mechanism of PP-g-St fibers.

homopolymerization of styrene monomers, so it is necessary to remove self-polymerized PS from PP-g-St by solvent extraction.

Characterization

The grafting percentage can be calculated using eq. (1):

$$\eta(\%) = (M_2 - M_1) / M_2 \times 100\% \quad (1)$$

where η (%) is the grafting percentage, M_1 (g) is the weight of PP fibers, and M_2 (g) is the weight of PP-g-St fibers. The Fourier transform infrared (FTIR) spectra of the original PP and grafted PP were studied using an FTIR spectrometer (Nexus 870, Thermo-Nicolet, Madison, United States). The samples were analyzed in the range 1900–650 cm^{-1} . The ^1H -NMR spectra of PP and PP-g-St were run on a Varian 400-MR NMR spectrometer (Palo Alto, United States). Solutions suitable for NMR measurements were prepared by dissolving ~ 10 mg of purified grafted polymers in deuterated tetrahydrofuran at 50 $^\circ\text{C}$. X-ray photoelectron spectroscopy (XPS) observations were conducted on a PHI 5000 VersaProbe (Ulvac-Phi, Kanagawa-ken, Japan), with an Al $K\alpha$ (1486.6 eV) monochromatic X-ray source operating at 20 mA current and 15 kV voltage. The pressure in the sample chamber was 10^{-8} torr. High-resolution scans were taken on the C1s with the energy resolution of 0.1 eV. The surface morphologies of PP fibers and PP-g-St fibers were observed by scanning electron microscopy (SEM; JSM-5900, Japan Electronics, Tokyo, Japan). Thermogravimetric analysis (TGA) of pristine PP and grafted PP-g-St fibers was performed on a Pyris1 TGA (Pyris 1 DSC, PerkinElmer, California, United States) from 25 $^\circ\text{C}$ to 700 $^\circ\text{C}$ under nitrogen at a heating rate of 20 $^\circ\text{C min}^{-1}$.

Adsorption of BTX

Adsorption capacities of PP-g-St fibers for BTX emulsion and solution in water were measured by the following procedure. First, approximately 1 g of the PP-g-St fiber sample was weighed and added to a 2 L beaker containing 1.5 L of deionized water, 20 mL of BTX, and 0.2 g of SDBS. After stirring for 1 h, it was left to stand for 2 h (25 $^\circ\text{C}$) to achieve a well-distributed and stable sample. The excess amount of BTX floating on the water surface was removed, and the initial concentration of the adsorption solution was tested. The maximum concentration of toluene and xylene in water was 483 and 350 mg/L, respectively. The emulsion and solution with lower concentration in water was obtained by adding less toluene and xylene. Then, PP-g-St fibers were fastened between annular metal grids and immersed in the mixture for 1 h with low-speed agitation using a magnetic rotor, to fully adsorb the BTX emulsion and solution in water. After that, the PP-g-St fibers were taken out and transferred to a separating funnel containing a certain amount of CCl_4 to extract the adsorbed BTX. The solution pH was adjusted to lower than 2, and the solution was oscillated for 3 min. The water in the extract of PP-g-St fibers was removed by adding 1.0 g of anhydrous sodium sulfate (an extra addition is needed if they all crystallize); then the solid was filtered and the volume for extraction was adjusted to 200 mL and maintained for 3 h. Finally, the concentration of BTX in the extract was determined using an oil concentration analyzer (ET1200, Euro Tech Co., Shanghai, China). The adsorption capacity of PP-g-St fibers can be calculated using eq. (2):

$$Q'(\text{mg/g}) = (0.2 \times C) / W \times 100\% \quad (2)$$

where Q' (mg/g) is the adsorption capacity of PP-g-St fibers for a BTX emulsion and solution in water, C (mg/L) is the concentration of BTX in the water, and W (g) is the original weight of PP-g-St fibers before adsorption. The test was repeated five times to obtain a mean value, and the adsorption capacity of the original PP fibers was measured for contrast at the same time.

The maximum adsorption capacity for BTX was also evaluated. About 1 g of PP-g-St fibers was weighed and then immersed in pure BTX or BTX emulsion and solution in water (20 mL, 25 $^\circ\text{C}$) to achieve the maximum adsorption. After holding the fiber corner in the air until the liquid on the surface stopped dripping, the fibers were weighed again. The maximum adsorption capacity of PP-g-St fibers was measured by eq. (3) (the test was repeated five times to obtain a mean value):

$$Q_s(\text{g/g}) = (W_1 - W_0) / W_0 \times 100\% \quad (3)$$

where Q_s (g/g) is the maximum adsorption capacity for pure BTX, W_1 (g) is the weight of PP-g-St fibers after maximum adsorption, and W_0 (g) is the original weight of PP-g-St fibers. The maximum adsorption capacity of PP fibers in pure BTX was also determined in a manner similar to the above procedure for comparison purposes.

Regeneration

Centrifugation, gravity extrusion, and solvent extraction were chosen as the regeneration methods for adsorption of pure toluene by PP-g-St fibers, and each cycle of regeneration and adsorption was repeated six times. The adsorption procedures were consistent with the methods described in the section "Adsorption of BTX." The conditions for each regeneration method were as follows: centrifugation at 4000 rpm for 10 min; gravity extrusion at -18.9 kPa for 10 min; solvent extraction, soaking in petroleum ether for 10 min; and then drying at 65 $^\circ\text{C}$ for 2 h. The regeneration adsorption efficiency can be calculated by eq. (4):

$$K = Q_r / Q_s \times 100\% \quad (4)$$

where K (%) is the regeneration adsorption efficiency, Q (mg/g) is the maximum adsorption capacity after regeneration, and Q_s (mg/g) is the original maximum adsorption capacity. The regeneration of PP-g-St fibers adsorbed with toluene emulsion and solution in water (with concentration of 170 mg/L) was investigated to evaluate the regeneration efficiency of toluene with lower concentration. Centrifugation was applied and regeneration-adsorption was repeated for six cycles.

RESULTS AND DISCUSSION

Effects of Reaction Parameters on Grafting Percentage

To study the extent of grafting of St on PP fibers, the influences of plasma reaction parameters on grafting percentage were investigated, as shown in Figure 2. First, the effect of input power of the plasma generator on grafting percentage is shown in Figure 2(A). It can be seen that by increasing the input power, the grafting percentage increased initially and then decreased. The grafting percentage reached a maximum value of 3.0% when the input power was 90 W. This was due to the degree of dissociation of the

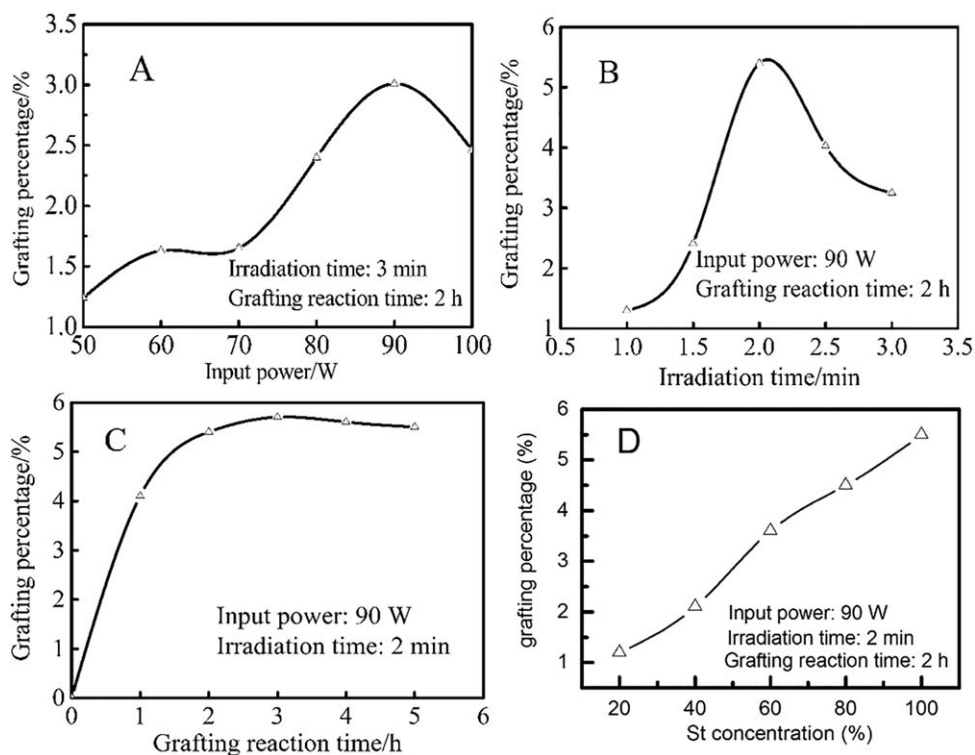


Figure 2. Effect of reaction parameters on grafting percentage of St: (A) effect of input power, (B) effect of irradiation time, (C) effect of grafting reaction time, (D) effect of St concentration.

plasma atmosphere and the increase in the velocities of active particles in the reactor with the increase in input power. In other words, active particles had higher density and kinetic energy, which increased the number of free radicals on the PP fiber surface and accelerated the *in situ* grafting reaction. However, when the input power exceeded 90 W, the grafting percentage decreased. This was because when the density and kinetic energy of active particles were too high, the surface of the PP fiber was bombarded and etched violently. This caused the fiber samples to lose weight, and so the grafting percentage decreased. For example, the weight of pristine PP fibers decreased from 6.102 g to 5.1852 g after plasma treatment under 60 W for 3 min, and the weight loss ratio is about 0.15%. The weight loss ratio increased to about 0.36% and 0.56% with plasma input power of 90 W and 100 W, respectively. Therefore, an optimal input power improves the grafting percentage, and in this paper the appropriate input power was found to be 90 W.

Second, the effect of irradiation time is shown in Figure 2(B). With the increase in irradiation time, the grafting percentage increased initially and then decreased. A maximum grafting percentage of 5.5% was reached when the irradiation time was 2 min. To some extent, the effects of extended irradiation time and increased input power were equivalent. Both these factors increased the number of free radicals on the surface of PP fibers and accelerated the grafting reaction. On increasing the irradiation time, the number of free radicals increased, and the grafting percentage also increased. When the irradiation time was long enough, the concentration of free radicals on the surface of PP fibers reached a saturation value, and the grafting percentage

was maximal. However, when the irradiation time was too long, the free radical concentration became too large, and the grafting percentage decreased. This was because the activation energies of adjacent free radicals were very low, and the coupling reaction between the free radicals by collision became more favorable, which decreased the grafting percentage.

Third, the influence of grafting reaction time on grafting percentage was studied, and the results are shown in Figure 2(C). The grafting percentage increased initially and then remained steady with extended grafting time. The grafting percentage reached a maximum value of 5.7% after 3 h. In the earlier period, the St monomers were fully in contact with the free radicals on the surface of PP fibers, which resulted in the rapid chain growth reaction of St. On extending the grafting time, the active sites on PP fiber surfaces decreased, and the chain growth reaction gradually slowed down and then terminated.

Lastly, the effect of St concentration is shown in Figure 2(D). St was dissolved in ethanol to obtain different concentrations in advance. The grafting percentage increased with increasing St concentration, and the maximum value was achieved with pure St. The grafting reaction occurred between St and PP, and the contact between active sites on PP and St was highly related to St concentration. A higher concentration of St made it feasible for the reaction to happen, while diluted St lowered the reaction efficiency.

Figure 3 shows the weight loss ratio of PP fibers irradiated with 90 W of power for different times. It can be seen that the weight loss of PP fibers was increased with increasing

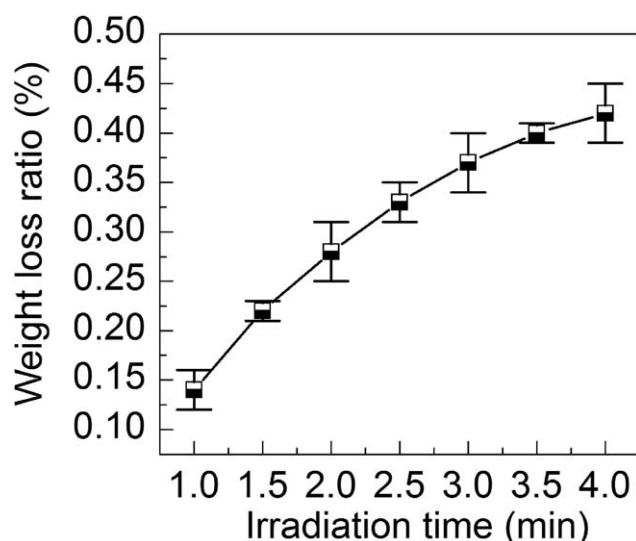


Figure 3. Weight loss ratio of PP fibers under various plasma irradiation times (90 W).

irradiation time from 1 min to 4 min. However, the weight loss ratio for treated PP is not remarkable, being just 0.28% for fibers irradiated for 2 min. In general, Ar plasma treatments involve two reactions, namely, a functionalization reaction as a result of inelastic collisions between the substrate and the activated ion species, and a chemical etching reaction to form small fragments due to decomposition of PP molecular chains.^{27,28} Though the active species including electrons and free radicals result in chemical etching on the surface of PP, the effect of the functionalization reaction is predominant during the plasma treatment.²⁸ So the weight loss of PP fibers under such a rapid and shallowly penetrating plasma treatment is negligible.²⁹

The grafting percentage of St on PP in this work is nearly equal to that induced by a photo irradiation grafting process under the same grafting period.¹⁹ Meanwhile, the grafting percentage is higher than that reported by Huang *et al.*²¹ (lower than 2.2%) and is lower than some previously reported results.^{20–25} The grafting percentage is linked to the specific reaction conditions, and it is difficult to make an actual comparison from values. In general, the grafting percentage of PP-g-St in this work is considerable given that no initiator and other polymerizers are used in the grafting reaction.

Characterization

It can be seen from Figure 4 that, compared to PP, the spectrum of PP-g-St showed almost the same absorption peaks, except for new peaks at 702 cm^{-1} and 758 cm^{-1} , whereas the peak at 1642 cm^{-1} became stronger. The peak at 1450 cm^{-1} was ascribed to the bending vibration peak of $-\text{CH}_2$, and the bending vibration peak of $-\text{CH}$ at 1380 cm^{-1} was also assigned to PP. The peaks at 702 cm^{-1} and 758 cm^{-1} could be attributed to the out-of-plane C—H bending vibrations of a monosubstituted benzene ring.³⁰ The peak at 1642 cm^{-1} was also ascribed to the vibrations of the benzene ring. PP-g-St fibers were extracted in acetone to remove the unreacted St monomers and its homopolymer, so these results clearly indicate the grafting of St on PP.

The $^1\text{H-NMR}$ spectra are shown in Figure 5. It can be seen from Figure 5(A) that PP shows the characteristic signals at 1.18 ppm for $[\text{dCH}_3]$ on PP molecular chains. On the other hand, PP-g-St shows, in addition to the characteristic PP signals, a strong signal at 6.91 ppm, which is related to the cyclo-benzene from St. This result verifies that St has been grafted onto the PP molecular chains.

XPS results are reported in Figure 6. C1s peaks characterizing the carbon atom compositions are shown on both the PP and PP-g-St fibers, and no other peaks are found. When examined at high resolution, carbon dominates the surface coverage at the 1s peak at $\sim 284.6\text{ eV}$ for PP fibers, corresponding to C—C/C—H functionalities. After grafting with styrene, the peak near $\sim 284.6\text{ eV}$ is slightly stretched compared with that of PP fibers. Furthermore, an additional peak is observed at $\sim 291\text{ eV}$, which can be assigned to the grafted styrene [Figure 6(B)].²¹

Figure 7 shows the images of PP fibers before and after plasma-induced grafting modification. Figure 7(B) reveals a smooth and flat surface of the PP fiber. After plasma irradiation [Figure 7(C)], the surface of PP fibers almost remained unchanged, which meant that besides creating active free radicals, plasma irradiation did not cause visible damage to the structure of the PP backbone and had little effect on the mechanical properties. The surface of PP-g-St fiber shown in Figure 7(D) is obviously covered with a corrugated layer. Due to the unchanged smooth and flat surface of the PP fibers after irradiation, it can be inferred that St chains were introduced on the surface of PP fibers after the grafting reaction. Moreover, the surface of PP-g-St fiber was rough and uneven, which could be attributed to the inhomogeneous grafting reaction. These depressions were uniformly distributed and had moderate depths, which contributed to the increased specific surface area of the fiber matrix.

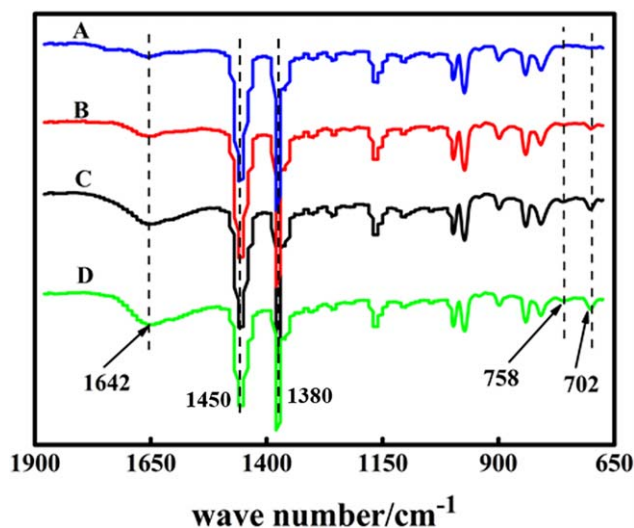


Figure 4. FTIR spectra of PP fibers before and after modification: (A) PP fibers; PP fibers irradiated by plasma at (B) 50 W, 3 min, and 2 h of grafting reaction; (C) 70 W, 3 min, and 2 h of grafting reaction; and (D) 90 W, 3 min, and 3 h of grafting reaction. [Color figure can be viewed at wileyonlinelibrary.com]

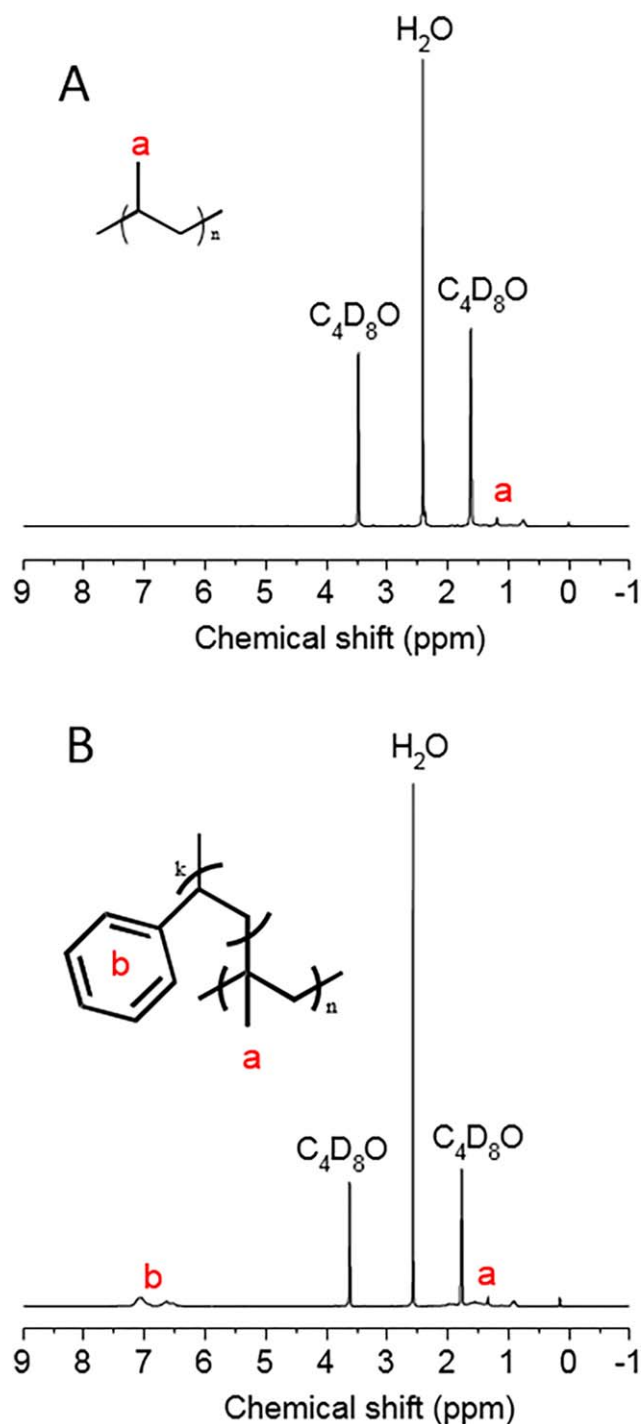


Figure 5. ^1H -NMR spectra of (A) PP and (B) PP-g-St. [Color figure can be viewed at wileyonlinelibrary.com]

This was beneficial for the adsorption process and had little effect on the mechanical properties.

TGA thermograms of the PP and the grafted PP-g-St are shown in Figure 8. The onset thermal degradation temperature of PP-g-St does not change too much compared with that of neat PP. The decomposition temperatures of PP-g-St fibers are slightly

increased from that of pristine PP fibers [Figure 8(A)]. The decomposition temperature increment is increased with increasing grafting percentage. Meanwhile, differential thermogravimetric (DTG) curves of PP and PP-g-St shown in Figure 8(B) reveal the same trend. The maximum degradation temperatures of PP, PP-g-St (3.0%), and PP-g-St (5.0%) are 477.9, 479.0, and 482.1 °C, respectively. It can be concluded that the slight increase of maximum degradation temperature for PP-g-St may be attributed to the St molecules bonded on the surface of PP fibers.

Adsorption of PP-g-St Fibers for BTX

The adsorption capacity of PP fibers and PP-g-St fibers for BTX emulsion and solution in water was studied, and the results are presented in Table I. It can be seen that, compared with PP fibers, the adsorption capacities of PP-g-St fibers for toluene and xylene emulsion and solution in water under different initial concentrations were significantly increased. The adsorption capacity of PP-g-St fibers toward toluene and xylene decreased with decreasing initial concentration. However, the adsorption ratio of toluene and xylene, namely the ratio between adsorbed toluene and xylene and the initial amount of toluene and xylene emulsion and solution in water, slightly decreased and remained relatively constant. The adsorption ratio of toluene was 84.57, 81.29, and 80.97% under concentrations of 483, 170, and 88 mg/L, respectively. The adsorption ratio of xylene was 97.76, 93.86, and 91.64% under concentrations of 350, 151, and 85 mg/L, respectively. So the results showed considerable adsorption capacity for PP-g-St fibers with low initial concentration of toluene and xylene. The adsorption capacity of fibers for benzene in water could not be tested accurately due to the volatile nature of benzene in water. The amount of benzene volatilized could be much higher than adsorbed over a limited time period.

Meanwhile, the maximum adsorption capacity of PP and PP-g-St fibers was also obtained. Figure 9 shows the maximum adsorption capacities of PP fibers and PP-g-St fibers for pure BTX. It was clear that, compared with PP fibers, the maximum adsorption capacities of PP-g-St fibers for pure benzene, toluene, and xylene showed considerable improvement, from 14.4 g/g, 13.5 g/g, and 10.9 g/g to 18.6 g/g, 16.8 g/g, and 13.4 g/g, respectively. As mentioned previously, this was because St was grafted onto the fiber surface and the specific surface area of fibers was increased. The adsorption capacity to BTX in this work is higher than in most of the previous research using inorganic materials as adsorbents.¹ As for organic adsorbents, the adsorption capacity in this work is comparable to or higher than that of some reported results.¹⁸ Compared with our previous work, the grafted PP-g-St shows an apparent increase of BTX adsorption compared to pristine PP fibers.⁷

The differences in increasing rates of adsorption capacities of PP fibers and PP-g-St fibers for pure BTX and BTX emulsion and solution in water have been discussed. First, the maximum adsorption capacities of PP fibers and PP-g-St fibers for pure BTX were large, and they could quickly adsorb BTX, more than 10 times their own weights. Meanwhile, their adsorption capacities and adsorption rates for BTX emulsion and solution in

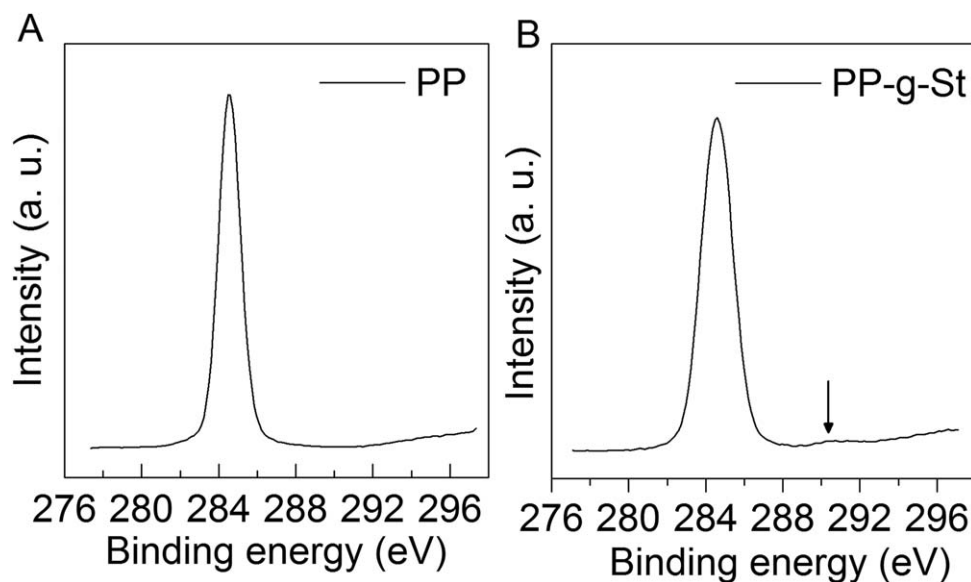


Figure 6. C1s spectra of (A) PP and (B) PP-g-St.

water were small. This was due to the web-like structures of PP fibers and PP-g-St fibers that allowed them to absorb large amounts of pure BTX quickly through their gaps by the capillary effect. However, for BTX emulsion and solution in water, due to the dispersion of BTX and the flow of water, BTX could not be properly absorbed inside the PP and PP-g-St fibers through the gaps.

Second, compared with pure BTX, the increasing rate of adsorption capacities of PP-g-St fibers for BTX emulsion and solution in water were larger than those of PP fibers. This could

be explained by surface adsorption and occlusion of PP-g-St fibers, which dramatically enhanced its adsorption capacity for BTX emulsion and solution in water. The solubility parameters of PP-g-St fiber and BTX were closer, which increased their affinities. BTX was tightly adsorbed on surfaces having St groups by van der Waals forces and was occluded by the swollen St chains on the fiber surface through solvation. Moreover, the surface of PP-g-St fibers was rough and had a larger specific surface area, which also increased the adsorption capacity of PP-g-St fibers for BTX emulsion and solution in water.

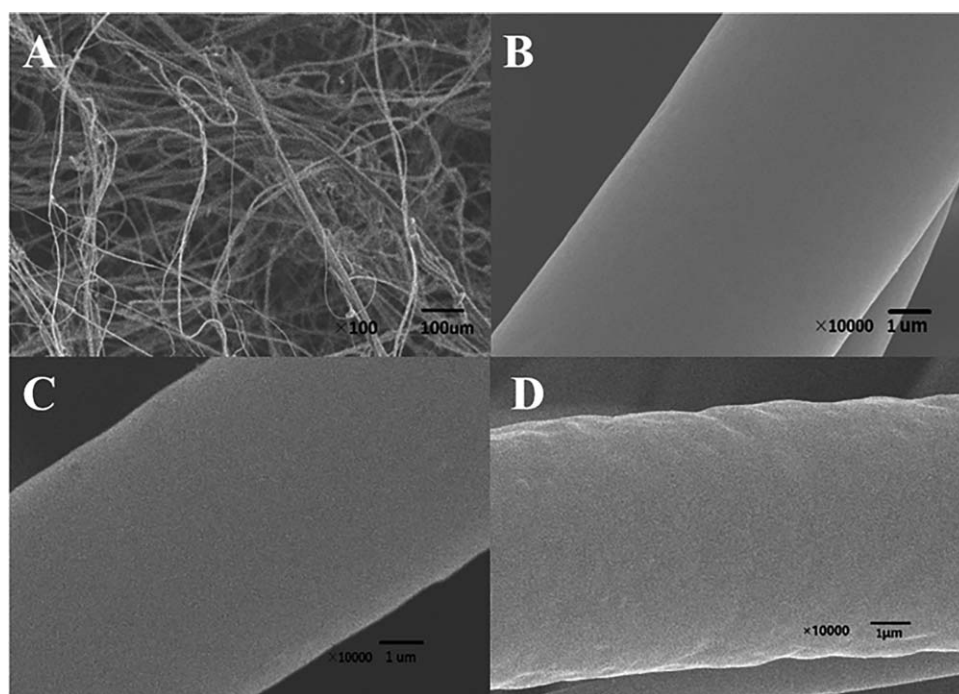


Figure 7. SEM images of (A) and (B) PP fibers, (C) PP fibers after plasma irradiation, and (D) PP-g-St fiber.

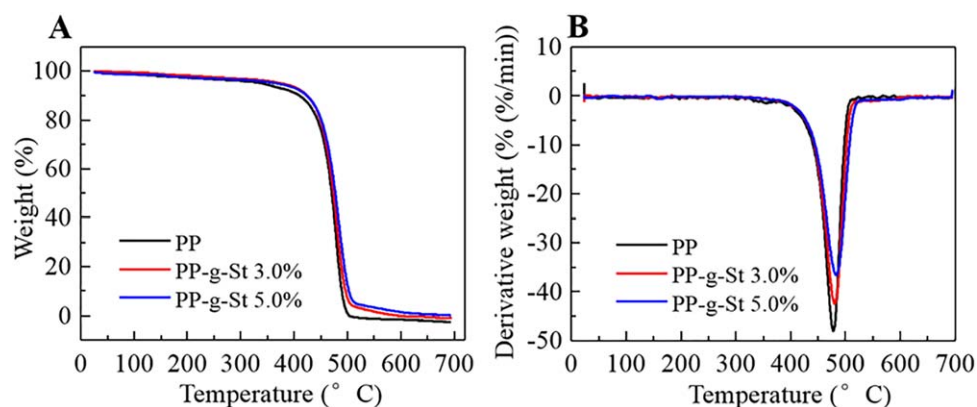


Figure 8. (A) TG decomposition curves and (B) DTG curves of PP and PP-g-St with grafting percentages of 3.0% and 5.0%. [Color figure can be viewed at wileyonlinelibrary.com]

Table I. Adsorption Capacities of PP Fibers and PP-g-St Fibers for BTX Emulsion and Solution in Water

Emulsified and dissolved BTX in water	Adsorption capacities of PP fibers (mg/g)	Adsorption capacities of PP-g-St fibers (mg/g)
Toluene (483 mg/L)	142.30 ± 1.27	612.70 ± 2.41
Toluene (170 mg/L)	42.25 ± 1.06	207.29 ± 2.04
Toluene (88 mg/L)	20.30 ± 0.85	106.88 ± 1.03
Xylene (350 mg/L)	115.45 ± 1.10	513.26 ± 2.73
Xylene (151 mg/L)	41.86 ± 0.98	212.59 ± 1.20
Xylene (85 mg/L)	22.61 ± 0.61	116.84 ± 1.00

Regeneration of PP-g-St Fibers

The desorption of adsorbed BTX and the regenerative efficiency of maximum adsorption of PP-g-St fibers are extremely important for practical applications. It can be seen from Figure 10

that the adsorption capacity of PP-g-St fiber for toluene was much higher. So, using pure toluene as an adsorbate (benzene was too volatile to calculate the regenerative adsorption efficiency), regenerative adsorption efficiencies of PP-g-St fibers by

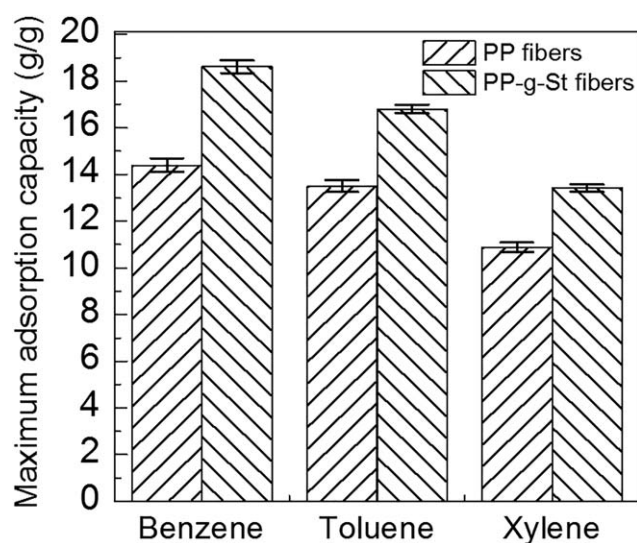


Figure 9. Maximum adsorption capacities of PP fibers and PP-g-St fibers for pure BTX.

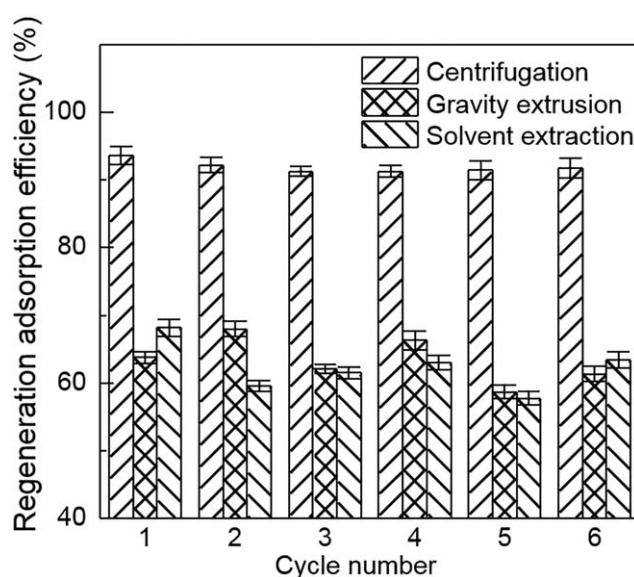


Figure 10. Regeneration adsorption efficiencies of PP-g-St fibers for three different methods.

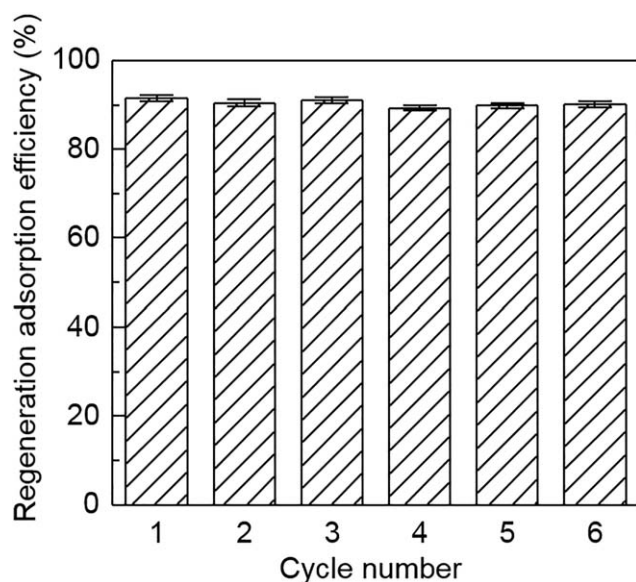


Figure 11. Regeneration adsorption efficiencies of PP-g-St fibers with toluene concentration of 170 mg/L.

regeneration methods like centrifugation, gravity extrusion, and solvent extraction were tested, and the results are shown in Figure 10.

It is evident from Figure 10 that, among the three regeneration methods, centrifugation was found to be the most efficient one. The high efficiency of this method could be attributed to the strength of the centrifugal force. BTX absorbed inside the web structure could be thrown out. Moreover, BTX absorbed and occluded in St chains by van der Waals forces could also get desorbed. But with the gravity extrusion and solvent extraction methods, desorption was difficult. After six cycles (adsorption–regeneration–adsorption) using centrifugation, the adsorption capacity remained stable, and the regeneration adsorption efficiency was still more than 90%. Most of the BTX absorbed on the surface of fibers was thrown out, but only a little amount of BTX that was occluded by the St chains could be desorbed.

Meanwhile, the regeneration adsorption efficiencies of PP-g-St fibers for toluene emulsion and solution in water was also studied, which are significant for practical use. It can be seen from Figure 11 that the regeneration adsorption efficiency of PP-g-St fibers toward toluene emulsion and solution in water was nearly 90% after six cycles, even with the concentration of 170 mg/L. This is consistent with the results in Figure 10, in which pure toluene was used. So it can be concluded that the regeneration efficiency was less affected by the concentration of toluene. The adsorption and regeneration of PP-g-St fibers of toluene emulsion and solution in water showed the prospect for practical applications of the fibers.

CONCLUSIONS

To improve the adsorption capacity toward BTX, St was grafted onto PP fibers, induced by Ar plasma irradiation without the addition of other chemicals. Characterization results from FTIR,

SEM, NMR, TGA, and XPS indicated that St was grafted on the surface of PP fibers. The optimum reaction parameters for grafting of St were input power of 90 W, irradiation time of 3 min, and reaction time for grafting of 3 h, to achieve a grafting percentage of 5.7%. Maximum adsorption capacities of PP-g-St fibers for pure BTX and BTX emulsion and solution in water were increased compared with that of pristine PP fibers. The adsorption capacities for benzene, toluene, and xylene showed a considerable improvement, from 14.4 g/g, 13.5 g/g, and 10.9 g/g to 18.6 g/g, 16.8 g/g, and 13.4 g/g, respectively. The grafted fibers exhibited good adsorption capacity over a large range of concentration of toluene and xylene, and the adsorption capacity was slightly decreased with decreasing concentration in water. The adsorption capacity of PP-g-St fibers was stable, and the regeneration adsorption efficiency was still more than 90% after six repeated cycles, using the centrifugation regeneration method. This study indicates that surface-functionalized PP fibers can be effective in cleaning up hydrocarbon pollutants and alleviating their harmfulness to the environment.

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